

Quiz1, MAT1575 Professor Chiu

ID: _____

Name: _____

- This quiz consists of 3 questions for a total of 10 points.
- You have 15 minutes to complete the quiz.
- Show all work and justify your answers.
- Wishing you success.

1. Evaluate the following indefinite integrals.

$$(a) \int \frac{3x^2 + 2}{x^2} dx = \int \left(\frac{3x^2}{x^2} + \frac{2}{x^2} \right) dx = \int 3 dx + \int \frac{2}{x^2} dx = 3x + \int 2 \cdot x^{-2} dx \\ = 3x + 2 \frac{x^{-1}}{-1} + C = 3x - 2x^{-1} + C = 3x - \frac{2}{x} + C$$

$$(b) \int (4\sqrt{x} + \sqrt[4]{x}) dx = \int \left(4x^{\frac{1}{2}} + x^{\frac{1}{4}} \right) dx = \int 4x^{\frac{1}{2}} dx + \int x^{\frac{1}{4}} dx = \frac{4x^{\frac{3}{2}}}{\frac{3}{2}} + \frac{x^{\frac{5}{4}}}{\frac{5}{4}} + C \\ = \frac{8}{3} x^{\frac{3}{2}} + \frac{4}{5} x^{\frac{5}{4}} + C$$

2. Find the general form for the antiderivative $F(x)$ of the given function $f(x)$.

$$f(x) = e^x - 3x^2 + \sin(x)$$

$$F(x) = e^x - \frac{3x^3}{3} - \cos(x) + C = e^x - x^3 - \cos(x) + C$$

Please turn over and finish the rest of the question

3. Let L_n denote the left-endpoint sum using n subintervals. Compute L_4 for the given function $f(x) = \frac{1}{x^2+1}$ on the interval $[-2, 2]$.

For L_4 , we slice $[-2, 2]$ into $n=4$ parts with $\Delta x = \frac{2 - (-2)}{4} = 1$

Then the partition is

$$P = \{-2, -2+\Delta x, -2+2\Delta x, -2+3\Delta x, -2+4\Delta x\}$$

$$= \{-2, -1, 0, 1, 2\} \quad \text{and}$$

$$L_4 = f(-2)\Delta x + f(-1)\Delta x + f(0)\Delta x + f(1)\Delta x$$

$$= \left(\frac{1}{(-2)^2+1} + \frac{1}{(-1)^2+1} + \frac{1}{(0)^2+1} + \frac{1}{(1)^2+1} \right) \Delta x$$

$$= \left(\frac{1}{5} + \frac{1}{2} + 1 + \frac{1}{2} \right) \cdot 1 = \frac{22}{10} = \frac{11}{5}.$$

End of this quiz.